

# Apple Inc. – Financial Analysis Report - 2025

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# A. Business Analysis – Apple Inc. (AAPL)

## 1. Company Overview

### • Brief History

Apple Inc. (NASDAQ: AAPL) was founded in 1976 by Steve Jobs, Steve Wozniak, and Ronald Wayne. Headquartered in Cupertino, California, Apple is a global leader in consumer technology, known for innovation in both hardware and software, initially focused on personal computers (Apple I, Apple II), revolutionized consumer tech with iPod (2001), iPhone (2007), and iPad (2010).

### • Mission & Strategic Direction

- Mission: Deliver innovative, high-quality technology products
- Strategy:
  - Premium product positioning
  - Strong ecosystem integration (iOS, macOS, services)
  - Focus on services growth (App Store, iCloud, Apple Music)

### • Major Milestones

- 2007: Launch of iPhone (major industry disruption)
- 2018: Became first \$1 trillion company
- 2020+: Expansion into services and wearables
- 2023+: Strong push into AI, chips (M-series), and ecosystem control

## 2. Current Operations

• **Main Products & Services:** Apple's core products include the iPhone, Mac, iPad, and Apple Watch. Its service portfolio features the App Store, Apple Music, iCloud, Apple TV+, and Apple Pay.

- Hardware: iPhone (largest revenue contributor) ; Mac, iPad ,Apple Watch, AirPods
- Services: App Store, Apple Music, iCloud, Apple Pay

### • Revenue Sources

- ~50%+ from iPhone
- Growing share from Services (high margin)
- Wearables segment increasing steadily

• **Geographic Markets:** Americas (largest), Europe, Greater China, Asia-Pacific, Global presence in 100+ countries

## 3. Macroeconomic Environment

### • Industry Conditions

Apple faces a **mature, competitive tech market** with strong demand for premium devices, regulatory pressures, and supply-chain challenges, while its ecosystem and services help sustain resilience.

- **Economic Factors Affecting Performance:** Apple's performance is influenced **by** global economic conditions, including:

- **Consumer spending power** – demand for premium devices depends on disposable income and economic confidence.
- **Currency fluctuations** – as a global company, foreign exchange rates impact revenue and profits.
- **Inflation & interest rates** – higher costs or tighter credit can reduce tech purchases.
- **Supply chain costs** – raw material prices and logistics affect margins.

- **Growth Strategy**

Apple primarily focuses on organic growth by innovating products, expanding services, and enhancing its ecosystem. It occasionally pursues acquisitions to strengthen technology capabilities (e.g., AI, AR/VR, or chip design), but M&A is secondary to its core strategy of in-house innovation and ecosystem expansion.

## 4. Industry & Competition

- **Major Competitors:** Samsung Electronics, Microsoft, Google, Amazon, Huawei, and Xiaomi.

- **Competitive Advantages:**

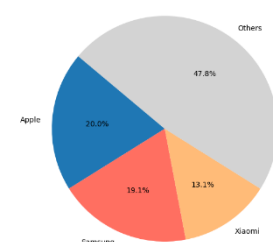
Apple Inc. has several competitive advantages, including strong brand loyalty, **an** integrated ecosystem combining hardware and software, high profit margins, custom silicon chips such as the M-series and A-series, and premium pricing power.

- **Approximate Market Share:**

According to IDC Worldwide Quarterly Mobile Phone Tracker (Final 2025): Apple 20.0%, Samsung 19.1%, Xiaomi 13.1% of global smartphone shipments.

(<https://www.idc.com/promo/smartphone-market-share/market-share> )

Approximate Global Smartphone Market Share — 2025 (IDC & Counterpoint)

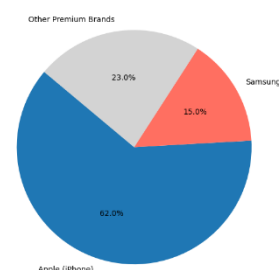


- **Premium segment:**

According to **Counterpoint Research's premium segment data** (devices priced over ~\$600), Apple continued to **dominate the global premium smartphone market**, capturing around: ~62% share of the global premium smartphone segment.

(<https://counterpointresearch.com/en/insights/global-premium-smartphone-sales-hit-record-h1-high-in-2025> )

Approximate Global Premium Smartphone Market Share — H1 2025 (Apple Dominance)



In conclusion, Apple's FY2025 success was characterized by a balanced performance between strong hardware sales — anchored by the iPhone — and

accelerating high-margin services revenue, reflecting a strategic shift toward a more diversified, ecosystem-driven business model.

## B. Ratio and Valuation Analysis

### 1. Profitability Ratios (FY2025)

| Ratio                    | Value   |
|--------------------------|---------|
| Gross Profit Margin      | 46.91%  |
| Operating Profit Margin  | 31.97%  |
| Net Profit Margin        | 26.92%  |
| Return on Assets (ROA)   | 31.18%  |
| Return on Equity (ROE)   | 151.91% |
| Earnings per Share (EPS) | 7.46    |

Apple shows exceptional profitability and efficiency with high margins (Gross 46.9%, Net 26.9%), strong asset utilization (ROA 31%), and impressive shareholder returns (ROE 152%, EPS 7.46). Performance is supported by premium pricing and a profitable ecosystem, though high leverage slightly increases financial risk.

### 2. Liquidity Ratios

| Ratio         | Value |
|---------------|-------|
| Current Ratio | 0.893 |
| Quick Ratio   | 0.859 |
| Cash Ratio    | 0.217 |

Apple's liquidity ratios are below 1 (Current 0.893, Quick 0.859, Cash 0.217), indicating short-term assets alone don't fully cover liabilities. However, strong and stable Free Cash Flow (~\$98.8B) ensures Apple can easily meet obligations, making low ratios manageable and not risky.

### 3. Additional Ratios

| Ratio                   | Value   |
|-------------------------|---------|
| P/E Ratio               | 34.07   |
| Dividend Payout Ratio   | 13.77%  |
| Debt-to-Equity Ratio    | 3.87    |
| Sustainable Growth Rate | 131.00% |

Apple shows **high growth and profitability** with a P/E of 34.1 and a Sustainable Growth Rate of 131%, indicating strong market expectations and capacity to expand. Dividend payout is low (13.8%), reflecting focus on reinvestment. Debt-to-equity is high (3.87), which **boosts ROE** but adds some financial risk.

### 4. CAPM and Cost of Equity

#### ❖ Definition of CAPM

CAPM stands for Capital Asset Pricing Model. It is a financial model that calculates the expected return of an asset (usually a stock or portfolio) based on its systematic risk (measured by Beta) relative to the overall market.

CAPM is essential in finance because it answers the key question: “What return should investors demand for the risk they are taking?”. Key applications of CAPM: Cost of Equity, Discounted Cash Flow (DCF) Valuation, Investment Decisions, Capital Budgeting Used to calculate WACC for evaluating investment projects, Portfolio Management

CAPM is a foundational model that links risk (Beta) to expected return. It is critically important in valuation because it provides the required rate of return used in almost all discounted cash flow models.

### ❖ Beta Calculation via Regression

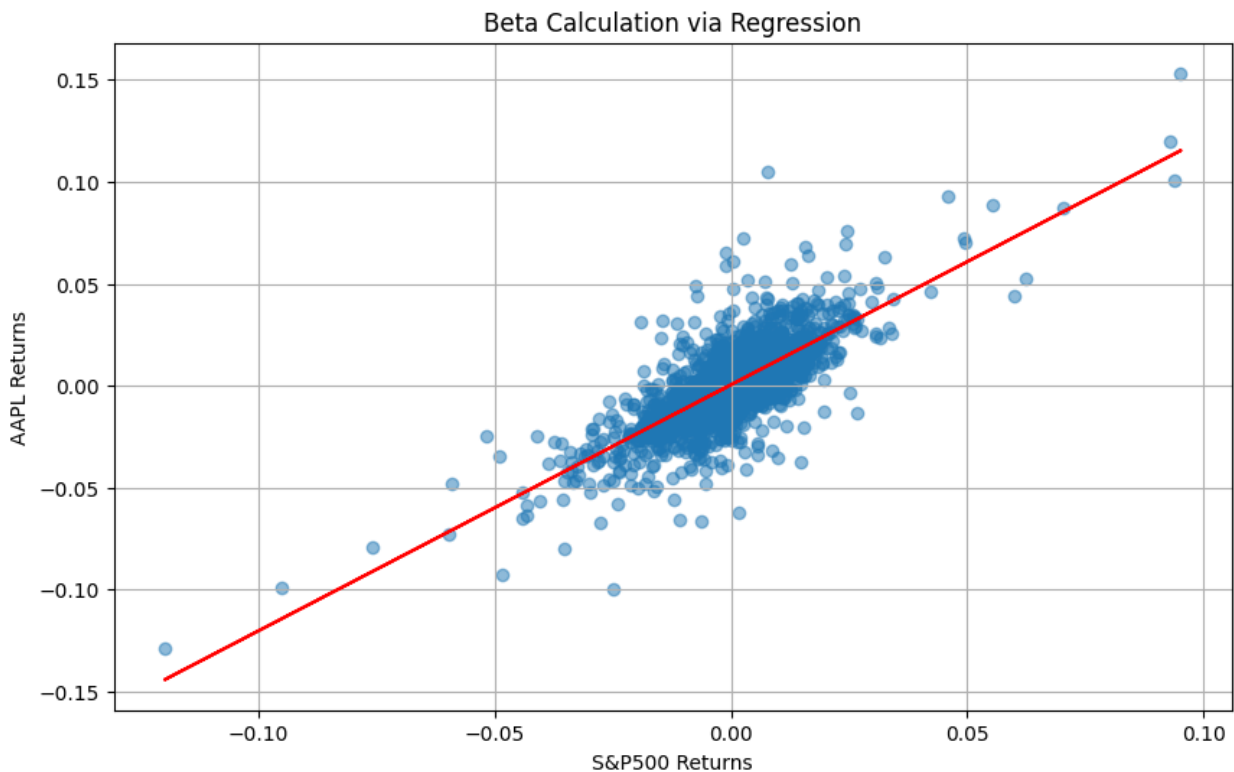
To measure Apple's market risk exposure, we performed a linear regression between the daily returns of Apple Inc. and the S&P 500 index.

The scatter plot shows a positive correlation between the two assets. The red line represents the regression line, and its slope (beta) is approximately 1.2, which indicates that Apple’s stock moves slightly more than the overall market.

Using 10 year of daily return data, we conducted a regression of Apple’s returns against the S&P500: Beta ( $\beta$ ) = 1.2

A beta greater than 1 suggests that AAPL is slightly riskier than the overall market. It offers higher potential returns in bull markets but also experiences larger drawdowns in bear markets. This is typical for large-cap tech/growth stocks like Apple.

This regression plot supports the CAPM-based risk assessment and enhances the understanding of Apple's market sensitivity.



### Expected Return (CAPM Formula)

Using the CAPM model, we calculate Apple's cost of equity based on data from the Apple 2025 Annual Report (Excel file), and from U.S. Department of Treasury – Daily Treasury Yield Curve Rates and yfinance:

- Risk-free rate = 0.0269 → 2.69%/year (10-year US Treasury yield - average 10 year start='2015-12-31', end='2025-12-31')
- Market return = 13.85% (S&P500 10-year average: start='2015-12-31', end='2025-12-31')
- Beta ≈ 1.2 (calculated from historical regression of AAPL vs. S&P 500 returns)

Applying the Capital Asset Pricing Model (CAPM):

$$ROE = R_f + \beta \times (R_m - R_f) = 2.69\% + 1.207 \times (13.85\% - 2.69\%) = 16.16\%$$

⇒ **This suggests that investors require approximately 16.16% annual return to compensate for Apple's market risk.**

### Application in making investment decision:

Given CAPM (required return) = 16.16%, we compare it with the stock's expected return to decide whether to buy.

Assume:

- ✓ Expected future price = \$290
- ✓ Dividend = \$4
- ✓ Fair value formula:

$$P_{today} = \frac{P_{future} + Dividend}{1 + CAPM} = \frac{294}{1.1616} \approx 253$$

### Decision rule:



- Current price < \$253 → Buy (undervalued)
- Current price ≈ \$253 → Hold
- Current price > \$253 → Not attractive

CAPM helps estimate a fair buying price based on risk. If Apple's market price is below this level, it may be a good investment opportunity.

## 5. Free Cash Flow (FCF):

Free Cash Flow (FCF) represents the cash a company generates from its operations after accounting for capital expenditures. It indicates the cash available to reinvest in the business, pay dividends, reduce debt, or pursue growth opportunities

### ❖ Calculate Free Cash Flow

For the fiscal year ended September 27, 2025, Apple Inc. generated \$111,482 million in cash from operating activities (CFO) and spent \$12,715 million on property, plant, and equipment (CapEx). Using the standard FCF formula:

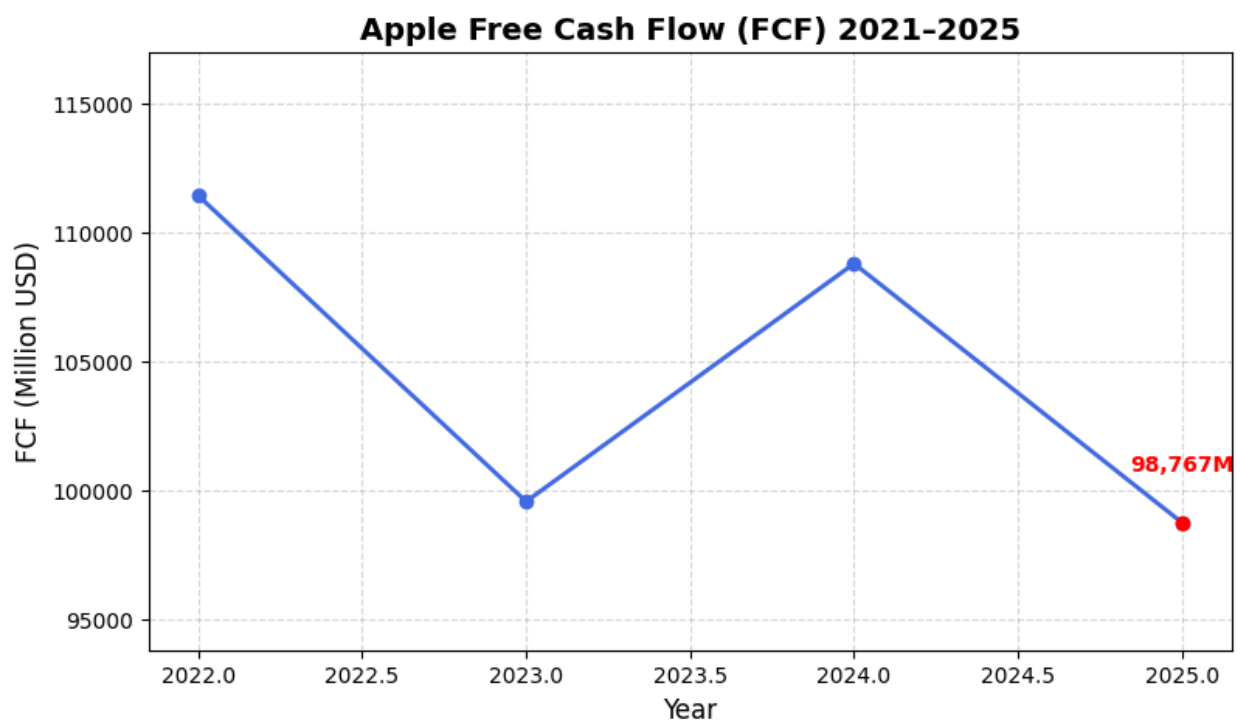
$$\text{FCF} = \text{Operating Cash Flow (OCF)} - \text{Capital Expenditures (CapEx)}$$

$$\text{FCF} = \text{CFO} - \text{CapEx} = 111,482 - 12,715 = 98,767 \text{ million USD}$$

Apple's Free Cash Flow for FY2025 is \$98,767 million.

### ❖ Analysis of sustainability and trends

Apple's Free Cash Flow over the past four years demonstrates strong and stable cash generation, supporting sustainable dividends, share buybacks, and strategic investments.



### ❖ Investor implications:

The substantial FCF suggests that Apple has significant financial flexibility, enabling it to:

- Pay dividends and repurchase shares, enhancing shareholder value.
- Invest in research and development, new products, or potential acquisitions.
- Maintain a strong balance sheet and reduce debt risk.

Investors can view Apple's FCF as a key indicator of financial health and long-term sustainability, reinforcing confidence in the company's ability to generate returns while funding growth initiatives.

## 6. Valuation (WACC & Intrinsic Value): WACC Sensitivity Analysis – Intrinsic Value per Share

The Weighted Average Cost of Capital (WACC) is estimated using the CAPM model based on a beta of **1.2**, a risk-free rate of **2.7%** (10-year Treasury), and a market return of **13.9%**, resulting in a WACC of **15.79%** as of December 2025.

Using this discount rate, the firm value is calculated by discounting Free Cash Flow (FCF) of **\$98,767 million**, yielding an intrinsic firm value of approximately **\$625,307 million**. Dividing by **15,015 million shares outstanding**, the intrinsic value per share is estimated at **\$41.65**.

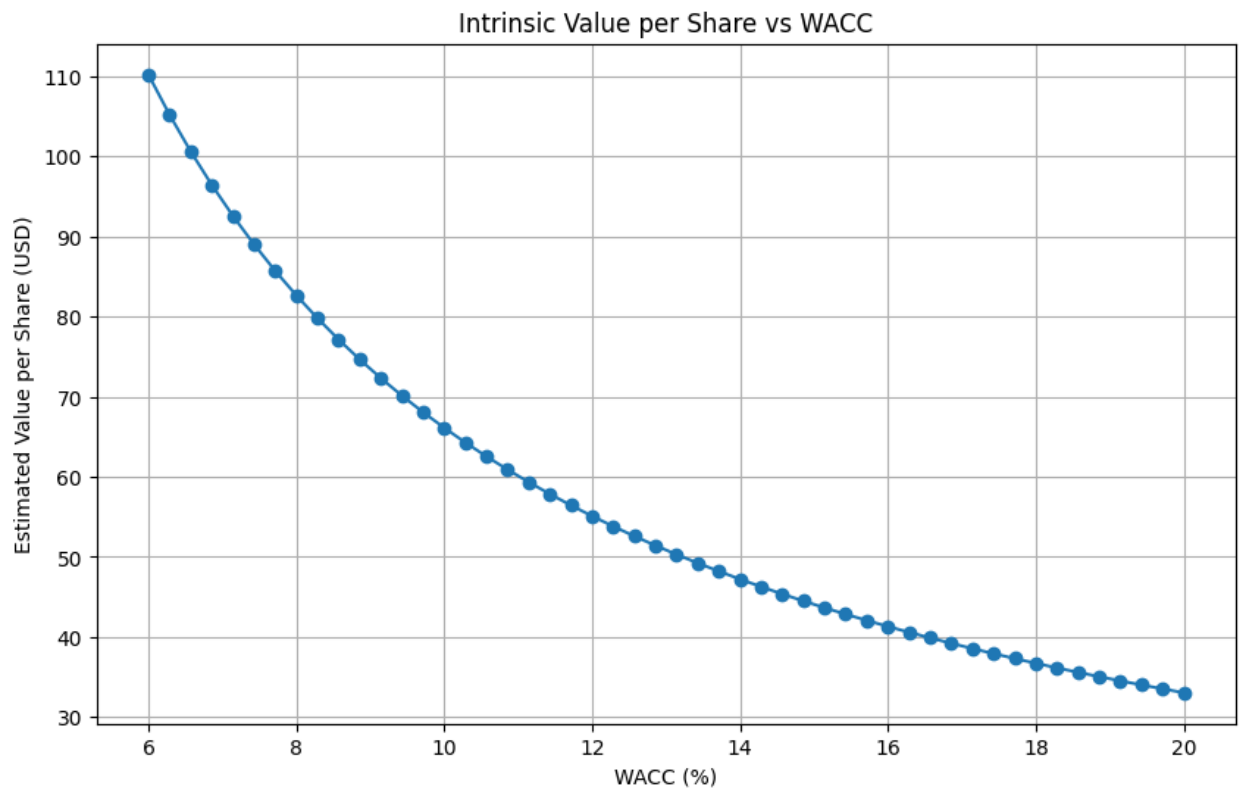
| WACC + Intrinsic Value |               |                                                    |
|------------------------|---------------|----------------------------------------------------|
| Item                   | Value         | Note                                               |
| FCF                    | 98,767        | Unit: million USD                                  |
| Cost of Equity (Ke)    | 16.08%        | CAPM: $R_f + \beta(R_m - R_f)$                     |
| Cost of Debt (Kd)      | 5.56%         | From 10-K Note 9 (weighted-avg effective rate)     |
| Tax Rate               | 15.61%        | Tax Provision / Pre-tax Income                     |
| Market Cap (E)         | 3,816,062     | Shares × Price per Share                           |
| Total Debt (D)         | 98,657        | CP + Term Debt (Current + NC)                      |
| Total Value (V = E+D)  | 3,914,719     |                                                    |
| <b>WACC</b>            | <b>15.79%</b> | $(E/V) \times K_e + (D/V) \times K_d \times (1-T)$ |
| Intrinsic Firm Value   | 625,307.20    | FCF ÷ WACC                                         |
| Shares Outstanding     | 15015         | Unit: million shares                               |
| <b>Value per Share</b> | <b>41.65</b>  | <b>Firm Value ÷ Shares</b>                         |

Compared to the current market price, this suggests that Apple may be significantly overvalued under the calculated WACC, highlighting the strong sensitivity of valuation to the cost of capital and market return assumptions.

This plot visualizes the inverse relationship between **WACC** and **intrinsic value per share**.

- As WACC increases, the estimated value per share decreases.
- Shows the impact of discount rate changes on valuation outcomes.

This chart makes it easier to interpret the sensitivity table visually.



## C. Technical Analysis

This section evaluates Apple Inc.'s stock trend using quantitative tools to assess short-term movement and volatility.

### 1. Stock Price Trend (Last 1 Year)

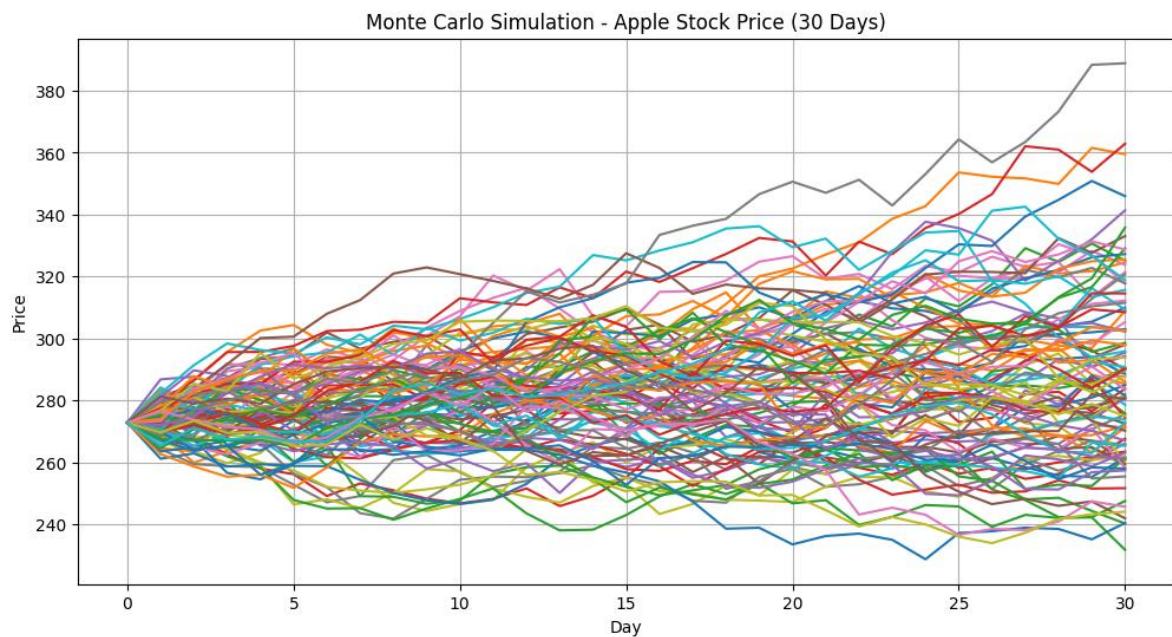
The historical daily closing price of Apple stock over the past 12 months was visualized using a line chart. Apple's stock shows a strong long-term upward trend over the past year, despite a mid-year decline and short-term consolidation. The recent price remains near historical highs, indicating strong market confidence but also potential overvaluation risks.



### 2. Monte Carlo Simulation (100 Simulations – 30 Days Forecast)

To estimate potential short-term price paths, we conducted a Monte Carlo simulation using Apple's recent daily returns. The simulation used the average daily return (mean) and standard deviation (volatility) to generate 100 random future paths over a 30-day period.

The Monte Carlo simulation suggests that Apple's stock is likely to move within a moderate range of \$270–\$310 over the next 30 days, with a slight upward bias but no strong trend. However, the wide spread of outcomes (approximately \$210 to \$360) indicates high short-term volatility and uncertainty, meaning both risk and potential return are significant.

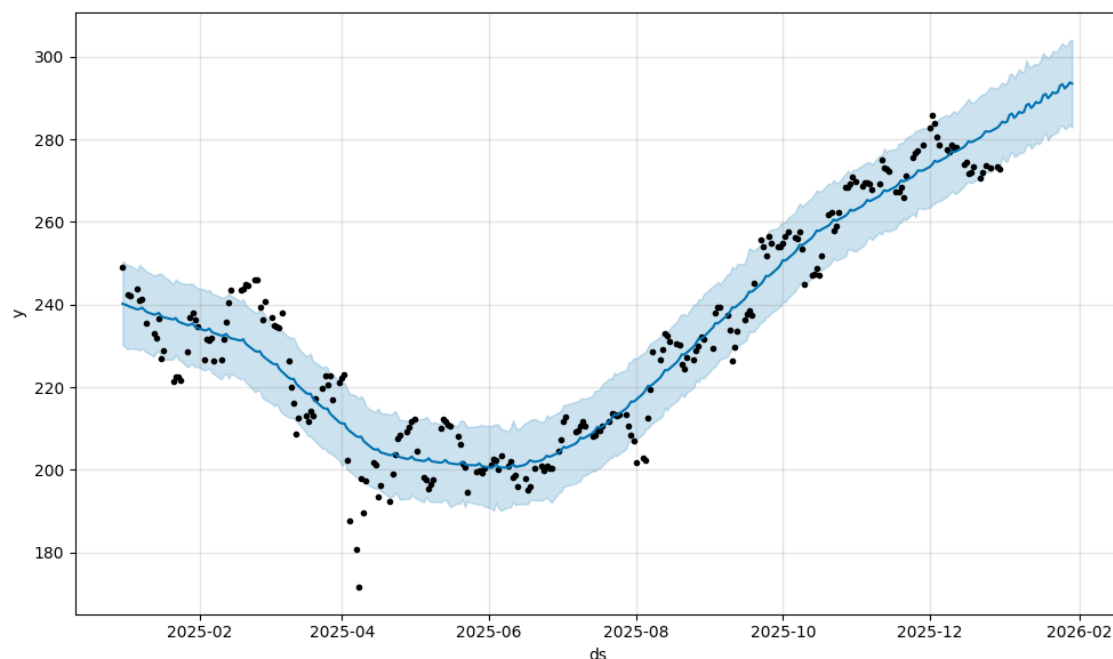


### 3. FB Prophet Forecast (Next 30 days)

We also applied Facebook Prophet, a time-series forecasting tool, to predict Apple's stock prices over the next 30 days. The model automatically detects trends and seasonality in time series data.

The Facebook Prophet forecast shows a clear upward trend after mid-2025, projecting Apple's price to continue increasing toward ~\$290–\$300. However, the confidence interval (shaded area) widens over time, indicating growing uncertainty in the forecast, which is typical as prediction horizons extend. The model suggests a bullish outlook, but the wide range implies that actual prices may vary significantly, so the forecast should be used with caution for decision-making. Refer to the Prophet forecast chart from the Jupyter Notebook.

#### FB Prophet Forecast Chart

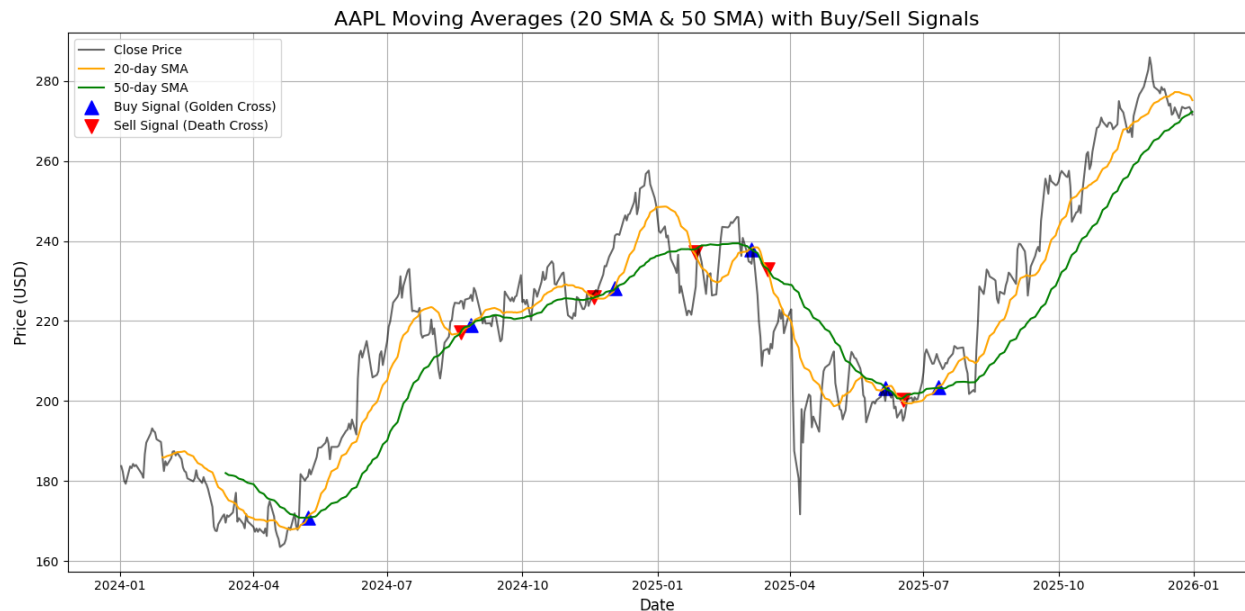


#### 4. Moving Averages & Trading Signals Analysis

This analysis uses yfinance to obtain Apple's stock data and applies 20-day and 50-day SMAs to identify trends.

Crossover points generate trading signals: a **Golden Cross** indicates a potential buy opportunity, while a **Death Cross** suggests a potential sell signal.

The purpose is to visualize price trends and support short-term trading decisions.



- ✓ Uptrend: When the 20-day SMA (orange) is above the 50-day SMA (green) → price tends to rise.
- ✓ Downtrend: When the 20-day SMA is below the 50-day SMA → price tends to fall.
- ✓ Golden Cross (▲ blue): Point where the 20-day SMA crosses above the 50-day SMA → potential buy signal.
- ✓ Death Cross (▼ red): Point where the 20-day SMA crosses below the 50-day SMA → potential sell signal.

## D. Investment Recommendation

Based on the financial ratios, intrinsic valuation, and technical analysis, **we recommend a HOLD strategy for Apple Inc. (AAPL).**

Apple Inc. demonstrates **strong financial fundamentals**, including:

- Stable and significant Free Cash Flow (~\$98.8B)
- Exceptional profitability and ROE (~151%)
- Solid ecosystem and diversified revenue streams (products + services), supporting long-term sustainability

Using a **Discounted Cash Flow (DCF)** analysis with a calculated **WACC of 15.79%** (derived from a Cost of Equity of 16.08% using CAPM with  $\beta = 1.2$ ,  $R_f = 2.7\%$ ,  $R_m = 13.9\%$ , and a Cost of Debt of 5.56% after 15.61% tax), **the intrinsic value per share is estimated at \$41.65**, based on free cash flow of \$98,767 million and 15,015 million shares outstanding. **This intrinsic value is below the current market price**, reflecting the sensitivity of the valuation to the relatively high discount rate.

Technical analysis confirms an uptrend with no major sell signals, indicating continued market confidence in the short term.

The Monte Carlo simulation further indicates stability with moderate upside, suggesting that Apple remains a strong long-term holding. And the 20 and 50-day Simple Moving Averages (SMA) indicate an uptrend with no clear sell signals. **Therefore, we do not advise immediate purchase, but recommend holding existing positions.**

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